

MULTI-REASONING AGENT DECISION WORKFLOW

A 3-agent Langflow pipeline that decomposes, decides, and red-teams a business decision before it reaches stakeholders.



The Problem

A single LLM prompt asked to “recommend a decision” skips the steps a rigorous reviewer would insist on.

01

No decomposition

Jumps straight to an answer without breaking the decision into evaluable tasks or naming missing information.

02

No forced commitment

Hedges with “it depends” instead of picking a defensible option under real constraints.

03

No self-critique

Never argues against its own recommendation — nothing plays devil's advocate before it reaches a stakeholder.

System Flow

Each arrow is a Langflow response → input_value edge carrying the full upstream Message. No shared memory store.



All three agents: OpenAI gpt-4.1 · tool-calling enabled · max_iterations = 15 · API key from OPENAI_API_KEY env var (not hardcoded)

This is the 3-agent spine of a broader 12-step design (Planner → Researcher → dual-lens Analyzers → Synthesis → Critic → Reviser → Confidence Scorer → Publisher) sketched in the project's prompt library — implemented here as a focused, working slice.

Key Decisions & Trade-offs

Decision	We Gain	We Risk
Sequential 3-agent chain, not one prompt	Each agent stays honest to a single role	3 LLM calls instead of 1 — more latency & cost
No shared memory / vector store	Simple to reason about — pure message passing	Critic never sees the raw brief, only the memo
Same model (GPT-4.1) for all 3 agents	Consistent behavior across roles	No per-role cost/capability tuning
max_iterations = 15 per agent	Hard safety cap on runaway tool loops	Caps reasoning depth on hard cases

Outcome: Captured Test Runs

Three NovaCart decision scenarios run end-to-end through the live Langflow Playground — the Critic agent's captured verdict on each:

Premium Subscription Tier

REVISE

Needs finance sign-off and customer appetite probing before launch.

Platform Migration

ESCALATE

No load testing or risk quantification — not ready for an informed decision.

Vendor Selection

REVISE

Timeline too aggressive — needs cost projections, a decision rubric, fallback plan.

None of the three runs defaulted to APPROVE — the Critic surfaced a distinct, scenario-specific gap each time.

Tech Stack

- **Langflow**

Visual flow builder — flow exported at last_tested_version 1.10.2

- **OpenAI GPT-4.1**

Backs all 3 agents via Langflow's Agent component, tool-calling enabled

- **Langflow Flow JSON**

The pipeline itself is defined and version-controlled as exported JSON

How to Run

Import codebase/03-Langflow-MultiReasoningAgentWorkflow.json into Langflow → set OPENAI_API_KEY → open Playground → paste a decision brief (samples in codebase/Prompts-Decision Briefs.pdf) → read the Planner → Analyzer → Critic output.